

PhoenixHRMS

SUMMARY

The Reporting & Analytics Module provides enterprise-grade reporting, business intelligence, workforce analytics, KPI monitoring, and executive decision support. The module transforms operational HR data into actionable insights.

VERSION

1.0

STATUS

Draft

MODULE ID

RPT

DOCUMENT TYPE

Functional Module Specification

RELATED DOCUMENTS

15-employee-management-module.md 16-attendance-management-module.md 17-leave-management-module.md 18-payroll-management-module.md 20-performance-management-module.md 21-learning-management-module.md

BUSINESS OBJECTIVES

Improve decision making Increase workforce visibility Support compliance Reduce manual reporting Enable predictive workforce planning Improve executive reporting

MODULE SCOPE - INCLUDED

Operational Reports Analytical Reports Executive Dashboards KPI Monitoring Custom Report Builder Workforce Analytics Predictive Analytics Data Warehouse Scheduled Reports AI Insights Real-Time Dashboards Data Exports

MODULE SCOPE - EXCLUDED

External Enterprise BI Platforms Accounting Analytics These integrate externally.

ANALYTICS ARCHITECTURE

Operational Systems → Data Collection Layer → Analytics Data Store → Data Warehouse → Analytics Engine → Dashboards → Reports → AI Insights

DATA SOURCES

Employee Module: Employee Data, Organization Structure, Demographics Attendance Module: Attendance Records, Overtime, Shift Data, Absenteeism Leave Module: Leave Balances, Leave Utilization, LOP Payroll Module: Salary Data, Payroll Costs, Tax Information Recruitment Module: Candidate Data, Hiring Funnel, Hiring Metrics

Performance Module: Goals, Reviews, Competencies, Succession Learning Module: Courses, Certifications, Skills, Learning Progress

ANALYTICS LAYERS

Layer 1: Operational Reporting Layer 2: Business Analytics Layer 3: Predictive Analytics Layer 4: AI Workforce Intelligence

REPORTING FRAMEWORK - REPORT TYPES

Operational Reports Compliance Reports Management Reports Executive Reports Analytical Reports Custom Reports

OPERATIONAL REPORTS

Employee Reports: Headcount, Employee Directory, New Joiners, Separations, Probation Status, Diversity Reports Attendance Reports: Daily Attendance, Monthly Attendance, Late Arrivals, Overtime, Absenteeism Leave Reports: Leave Balances, Leave Utilization, Carry Forward, Encashment, LOP Payroll Reports: Payroll Register, Tax Reports, PF Reports, ESI Reports, Department Cost Reports

RECRUITMENT REPORTS

Open Positions Candidate Funnel Source Effectiveness Offer Acceptance Rate Recruiter Productivity Time To Hire Cost Per Hire

PERFORMANCE REPORTS

Goal Completion Performance Distribution Competency Gaps Talent Matrix Succession Reports PIP Reports

LEARNING REPORTS

Course Completion Learning Hours Skill Growth Certification Status Compliance Training Learning Effectiveness

ASSET REPORTS

Asset Inventory Asset Utilization License Usage Maintenance Reports Asset Recovery Warranty Reports

EXECUTIVE DASHBOARDS

CEO Dashboard: Headcount, Attrition, Hiring Status, Workforce Costs, Performance Trends, Leadership Pipeline HR Dashboard: Employee Metrics, Attendance Metrics, Leave Metrics, Engagement Metrics, Compliance Metrics Finance Dashboard: Payroll Costs, Benefits Costs, Tax Liability, Department Costs, Workforce ROI

KPI FRAMEWORK

Workforce KPIs: Headcount, Attrition Rate, Retention Rate, Absenteeism Rate, Employee Growth Recruitment KPIs: Time To Hire, Time To Fill, Cost Per Hire, Offer Acceptance Rate Performance KPIs: Goal Achievement, Review Completion, Promotion Rate, PIP Success Rate Learning KPIs: Learning Hours, Completion Rate, Certification Rate, Skill Growth

WORKFORCE ANALYTICS

Demographic Analytics: Gender Distribution, Age Distribution, Tenure Distribution, Location Distribution, Diversity Metrics Workforce Composition: Department Distribution, Role Distribution, Employment Type Distribution, Grade Distribution

ATTRITION ANALYTICS

Metrics: Voluntary Attrition, Involuntary Attrition, Department Attrition, Location Attrition, Manager Attrition
Analysis: Trend Analysis, Root Cause Analysis, Risk Indicators

RECRUITMENT ANALYTICS

Hiring Funnel Candidate Conversion Source Performance Recruiter Effectiveness Hiring Velocity Quality Of Hire

PAYROLL ANALYTICS

Payroll Cost Trends Department Cost Analysis Overtime Cost Analysis Tax Analysis Employer Contributions
Compensation Trends

PERFORMANCE ANALYTICS

Performance Distribution High Performer Identification Competency Gaps Promotion Readiness Succession Coverage

LEARNING ANALYTICS

Skill Gap Analysis Learning Effectiveness Course Effectiveness Certification Trends Development Progress

PREDICTIVE ANALYTICS

Workforce Forecasting: Headcount Forecast, Hiring Forecast, Attrition Forecast, Skill Demand Forecast, Payroll
Forecast Risk Analytics: Flight Risk, Skill Shortage Risk, Succession Risk, Compliance Risk

AI WORKFORCE INTELLIGENCE

AI Insights: Attrition Predictions, Promotion Recommendations, Learning Recommendations, Workforce
Optimization, Succession Insights, Compensation Insights Governance: Recommendations only. Human review
mandatory.

DATA WAREHOUSE

Purpose: Central analytics repository. Characteristics: Historical Data, Aggregated Data, Optimized Queries, Analytics
Processing Retention: Unlimited Historical Analytics, Configurable.

CUSTOM REPORT BUILDER

Features: Drag & Drop Builder, Field Selection, Filters, Sorting, Grouping, Calculated Fields, Charts Output Formats:
Table, Chart, Pivot, Summary

VISUALIZATION FRAMEWORK

Chart Types: Bar, Line, Pie, Donut, Area, Heatmap, Scatter Plot, Tree Map, Gauge Dashboard Components: Widgets,
Tables, KPIs, Charts, Cards

REAL-TIME ANALYTICS

Real-Time Metrics: Employees Online, Attendance Today, Pending Approvals, Recruitment Pipeline, Learning
Activity Refresh Frequency: Configurable, Default 60 Seconds

SCHEDULED REPORTS

Features: Daily Reports, Weekly Reports, Monthly Reports, Quarterly Reports, Custom Schedules Delivery Methods: Email, Portal, File Export, Cloud Storage

DATA EXPORT

Formats: Excel, CSV, PDF, JSON, XML Features: Scheduled Export, Bulk Export, Filtered Export

ANALYTICS APIS

Reports: GET /reports, POST /reports Dashboards: GET /dashboards, POST /dashboards KPIs: GET /kpis Analytics: GET /analytics Forecasts: GET /analytics/forecasts

DATABASE TABLES

Reporting Domain: reports, report_definitions, report_executions Dashboard Domain: dashboards, dashboard_widgets Analytics Domain: analytics_metrics, analytics_snapshots Forecasting Domain: forecast_models, forecast_results Data Warehouse Domain: fact_tables, dimension_tables, aggregation_tables

PERMISSIONS MATRIX

report.view report.create report.edit report.export dashboard.manage analytics.view analytics.manage forecast.view executive_dashboard.view

SECURITY REQUIREMENTS

Row-Level Security Tenant Isolation Sensitive Data Masking Salary Data Restrictions Audit Logging RBAC Enforcement

AUDIT REQUIREMENTS

Report Created Report Modified Report Executed Dashboard Modified Analytics Accessed Export Generated Forecast Executed Retention: 7 Years Minimum

INTEGRATIONS

Employee Module Attendance Module Leave Module Payroll Module Recruitment Module Performance Module Learning Module Asset Module Document Module Notification Module External BI Platforms

DASHBOARD WIDGETS

Headcount Attendance Rate Leave Utilization Payroll Cost Open Positions Goal Completion Learning Hours Asset Utilization Attrition Risk

PERFORMANCE REQUIREMENTS

Dashboard Load: < 3 Seconds Report Generation: 100,000 Records, < 10 Seconds Analytics Queries: < 5 Seconds

COMPLIANCE REQUIREMENTS

GDPR SOC 2 ISO 27001 Data Retention Auditability Regulatory Reporting

FUTURE ENHANCEMENTS

Version 2 Self-Service BI Natural Language Analytics Conversational Reporting AI Dashboard Builder Predictive Workforce Planning Advanced Data Science Workbench Organizational Network Analytics Digital Twin Workforce Model

SUCCESS METRICS

Report Accuracy: 100% Dashboard Availability: 99.99% Forecast Accuracy: > 85% Executive Adoption: > 90%

CONCLUSION

The Reporting & Analytics Module provides a comprehensive enterprise-grade workforce intelligence platform that transforms operational HR data into actionable business insights. The module integrates reporting, dashboards, KPIs, predictive analytics, AI-powered workforce intelligence, and executive decision support into a unified analytics ecosystem while maintaining scalability, performance, security, compliance, and tenant isolation.